Project RoyalDev UseCases:

Title	UC-AUTH-01: Register an account
Actors	ACT-01 (Registered user)
Preconditions	User has navigated to the web application URL and is not authenticated.
Basic Flow	1. User selects “Sign Up” from login page 2. User provides required details (e.g., email address, full name, preferred password). 3. Front-end validates passwords match 4. Backend validates input, checks for existing user conflicts, and applies a secure password hashing function. 5. Backend hashes passwords and creates the user record in the relational database. 6. System shows ‘account created’
Alternate Flow	1. Invalid / Duplicate Registration Details: System flags invalid input, rendering user-friendly validation messages without exposing system architecture details. 2. Invalid Credentials: System alerts the user of incorrect email/password combinations 3. Username taken: System shows username already exists
Postconditions	User account is created and stored in database; user is authenticated and redirected to Dashboard.
Open Issues	Redirect user to home page

Title	UC-AUTH-02: Sign in
Actors	ACT-01 (Registered user)
Preconditions	User is at the sign in page with registered credentials.
Basic Flow	1. User enters registered credentials. 2. System validates credentials against the database. 3. User is granted access and directed to the Dashboard.

Alternate Flow	Invalid Credentials: System alerts user of bad credentials and rate-limits login attempts.
Postconditions	User session established and user redirected to Dashboard.
Open Issues	Limit login rates

Title	UC-AUTH-03: Persistent User Sessions
Actors	ACT-01 (Registered user)
Preconditions	User has previously completed authentication and holds a valid refresh token or secure cookie session.
Basic Flow	1. User reopens or reloads the web application in a browser. 2. System checks secure client storage for active session tokens. 3. Backend validates session tokens and returns active user state seamlessly without requesting credentials. 4. User is restored to their authenticated dashboard view.
Alternate Flow	Expired / Invalid Session: System clears invalid session tokens and redirects user to the login screen with a session timeout notification.
Postconditions	Active session token is validated and user access is restored to authenticated state.
Open Issues	12-hour limit and add 30-minute timeout if inactive

Title	UC-AUTH-04: Sign Out
Actors	ACT-01 (Registered user)
Preconditions	User is signed in with an active session.
Basic Flow	1. User selects Log out from the navigation. 2. Frontend sends a logout request to the backend. 3. Backend marks the user's session as revoked in the database. 4. Backend clears the session cookie in the browser. 5. System redirects the user to the login page.

Alternate Flow	1. Session already expired or invalid: logout still succeeds; the cookie is cleared and the user is redirected to the login page. 2. Server unreachable: the user is still redirected to the login page; the server-side session expires automatically after 12 hours.
Postconditions	Session is revoked and the cookie is removed; protected pages require signing in again(needs valid cookie)
Open Issues	

Title	UC-BUD-01: Create a monthly budget
Actors	ACT-01 (Registered user)
Preconditions	User is successfully authenticated and has no current month budget
Basic Flow	1. User navigates to the Budget section of the dashboard. 2. User enters expected budget for month and plan 3. User allocates expected spending for needs and wants, $0 is accepted 4. User saves budget created 5. Backend validates input and System updates the budget and allocations 6. System displays new budget information
Alternate Flow	1. Invalid Input: System highlights input fields with an error notification and prevents form submission until corrected. 2. Budget Already Exists: System shows message to user, refresh and load budget 3. Session Expired: System loads log in page
Postconditions	Monthly category budget allocations are saved to database and displayed with active progress
Open Issues	

Title	UC-BUD-02: Budget Progress
Actors	ACT-01 (Registered user)

Preconditions	User is signed in and has a current budget for current month
Basic Flow	1. User selects Budget from the navigation. 2. System loads the month's budget with allocations, categories, and transactions. 3. System displays the plan: expected budget, chosen plan, needs and wants targets with expected spending, and savings target with expected savings. 4. System displays actual spending per category: amount spent and amount left. 5. System displays a summary: income received, spent so far, and projected savings.
Alternate Flow	1. No budget for the current month: system shows the budget creation form (UC-BUD-01). 2. Category overspent: the amount left is shown in red as a negative value. 3. Income below expected: system shows a warning with the worst-case savings.
Postconditions	User able to see progress, no data changed
Open Issues	

Title	UC-DASH-01: Dashboard
Actors	ACT-01 (Registered user)
Preconditions	User is signed in with an active session.
Basic Flow	1. User signs in or selects Home from the navigation. 2. Frontend requests the user's profile, categories, current month's budget, current month's transactions, and 5 most recent transactions. 3. Backend verifies the session and returns only that user's data. 4. System calculates projected savings, needs and wants progress, and spending so far. 5. System displays the monthly budget summary and recent transactions.
Alternate Flow	1. No budget for the current month: system shows a message with a link to create a budget.

	2. No transactions: system shows a message with a link to add a transaction. 3. Income below expected: system shows a warning with projected savings if the missing income doesn't arrive. 4. Session expired: system redirects to the login page.
Postconditions	Dashboard view presented to user on home page, no data change
Open Issues	

Title	UC-FIN-01: Log simulated transaction
Actors	ACT-01 (Registered user)
Preconditions	User is signed in with active session
Basic Flow	1. User selects Transaction Page 2. User enters a valid amount, selects income OR expense, category, and notes 3. Frontend validates input fields 4. User 'adds' transaction 5. Backend verifies category matches type(income or expense) and saves transaction 6. Transaction is shown history, and budget progress/tracking reflects this
Alternate Flow	1. Invalid input: System displays error message 2. Session Expired: Redirects to login page
Postconditions	Transaction log is written to the database and budget updated
Open Issues	Edit/delete transactions

Title	UC-FIN-02: Transaction history
Actors	ACT-01 (Registered user)
Preconditions	User is signed in with active session

Basic Flow	1. User selects Transactions from the navigation. 2. System loads the user's transactions for the current month. 3. System lists income and expenses separately, newest first, showing category, notes, date, and amount. 4. System displays monthly totals for income, expenses, and net activity. 5. User selects the previous or next month; the system loads and displays that month.
Alternate Flow	1. No transactions in the selected month, system shows "No transactions this month." 2. Long notes: shown on one line, with the full text on hover. 3. Session expired: system redirects to the login page.
Postconditions	No data is changed; the user sees the selected month's transactions.
Open Issues	

Title	UC-AI-01: Generate AI budget insights
Actors	ACT-01 (Registered user), ACT-02 (AI/LLM Service)
Preconditions	User has created at least one category budget and logged relevant transaction data.
Basic Flow	1. User accesses the assistant chat interface. 2. Backend compiles anonymized spending and budget data into a structured system prompt for the AI endpoint. 3. ACT-02 (AI/LLM Service) processes the context and returns actionable, customized recommendations. 4. Frontend displays formatted insights cleanly on the user interface.
Alternate Flow	AI Service Timeout/Unavailable: System captures API communication failures gracefully and presents a fallback message advising the user to try again shortly.
Postconditions	Formatted AI recommendations are displayed cleanly on user interface

Title	UC-PROF-01: Display Profile Info and Settings
Actors	ACT-01 (Registered user)
Preconditions	User is authenticated and logged into the application.
Basic Flow	1. User navigates to the Profile & Settings page. 2. System retrieves user details (e.g., name, email, notification preferences) and displays them on the interface. 3. User updates preference settings or profile information and submits the changes. 4. Backend validates the modifications and persists updated records to the database.
Alternate Flow	Invalid Input: System highlights invalid fields and displays relevant error messages preventing submission.
Postconditions	User profile and setting modifications are validated and stored in the database.
Open Issues	